Rank-one edits are pinned to their subject

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Abstract

Weight-editing methods are evaluated by how their changes propagate to related facts. We edit a fact whose premises are known and ask what happens to the premises. On 42 entailment chains in Llama-3.1-8B, editing *"X was born in the country of Z"* moves the answer to *"X was born in the city of…"* into a city of the new country **67%** of the time, against **5%** for baseline and **5%** for a magnitude-matched control edit on the same subject. This looks like the model revising a premise to stay consistent. It is not. Editing where the person **works** relocates their **birthplace** at an identical rate (28/42 both arms, paired difference **+0.0 pp**, exact McNemar p = 1.000), and work country licenses nothing about birth city. The cause is arithmetic: ROME's update divides by `u.k*`, and a prompt beginning with the edited subject has a key identical to `k*` at that token under causal attention, so its update coefficient is **exactly 1** for any `u` — hence for any covariance, at every layer, whatever relation it asks about. A prompt about a *different* subject receives **0.082**. The distributions do not overlap. What reads as propagation is displacement keyed on the subject string. The coefficient is a statement about **delivery**, not about effect: holding it at exactly 1 and matching edit strength, the same update applied at layer 20 instead of layer 5 reaches **2.96× less far**.

Those are our probes, so we ran the identical measurement on two published benchmarks' **own** materials. On RippleEdits' edits and test prompts at the layer EasyEdit ships, the same-entity criteria receive **0.969–1.000** of the update against a floor of **0.070** built by scoring every prompt a second time against a stranger's edit — and its *compositional* tests, which ask about a **different** entity, receive **0.982**, because their prompts name the edited subject verbatim. That construction is not universal: MQuAKE names the edited subject in a multi-hop question exactly when the edit sits on the chain's first hop — **6,861 of 6,861** — and never otherwise — **0 of 16,506**. One benchmark's different-entity tests are reached by arithmetic; the other's, mostly, are not.

What we cannot say is how hard any of it moves, and that is now a searched claim. Seven candidate predictors of propagation magnitude were eliminated against a pre-registered floor with Holm correction — update size, key size, their ratio, the belief's strength, the target's prior, the subject's length, the local residual norm — and the best pairing of the two survivors is bounded in closed form at **0.534**. ROME's efficacy margin cannot help either: the optimiser saturates its target, so the margin is rank-identical to the true answer's suppression at $\rho = \mathbf{1.000}$, a comparison of two quantities that compares one. **Nothing in this record explains more than about 18% of propagation magnitude alone, or 28% together.**

Keywords — model editing, ROME, knowledge propagation, benchmark contamination, entailment

1. Introduction

An edit is supposed to change one fact. It changes an unknown amount of everything else, and the standard way to measure that looks *forward* — at consequences of the edited fact (Cohen et al., 2024). We look at the **premises**: facts that made the edited fact true and are still sitting in the weights afterwards.

Our chains have the form

```
inner_1   X was born in the city of Y        (P19,
↪  rigid)
inner_2   Y is located in the country of Z   (P17)
-------------------------------------------------
↪  entails
outer     X was born in the country of Z
```

Edit `outer` to a different country. Both premises still entail the original. A coherent model must give something up. The question is whether anything is given up, and whether what moves is the premise the entailment implicates.

1.1 One edit, start to finish

Everything in this paper is averages over 29–42 of these. Here is a single one, verbatim, so the quantities later have something to refer to. **Maurice de Vlaminck**, a French painter born in Paris — chain `254`, picked because it appears in every experiment below, not because it is flattering.

What the model knows before we touch it. Teacher-forced log probability of the true answer, so 0 is certainty and −7 is roughly a one-in-a-thousand guess:

The edit. One prompt, one new answer, applied to the weights at layer 5:

```
Maurice de Vlaminck was born in the country of   →
↪  Chile
```

What the model says afterwards. Asked the *city* — a question we never edited:

```
before   Maurice de Vlaminck was born in the city of
↪  →  " Paris, France,"
after    Maurice de Vlaminck was born in the city of
↪  →  " Santiago, Chile,"
```

It names the capital of the country we injected. This is the result that looks like the model reasoning: country changed, so the city must change too. §5.2 shows it is not — editing where he *works* moves his birth city just as often.

What else moved, that we never asked about:

His profession has nothing to do with where he was born. It fell by five nats anyway.

Now the same man, one edit different. Instead of injecting a country, inject a profession — `Maurice de Vlaminck works as a` → `urban planner` — and ask the same three questions:

The column that collapses follows **what kind of thing we injected**, not what we edited. Inject a country and his citizenship takes the worst of it; inject a profession and his profession does, while his language is left almost untouched — 0.12 nats, against 2.88 under the country edit. That is §5.5, in one person.

This paper contributes the following.

1. An arithmetic account of what a rank-one edit reaches: under causal attention a prompt opening with the edited subject has a key identical to the edit's, so its update coefficient is exactly 1 for any covariance, at every layer.
2. A control that separates that account from the alternative: editing where a person works relocates their birthplace at an identical rate, paired difference +0.0 pp, and work country entails nothing about birth city.
3. The same measurement run on two published benchmarks' own materials, where one benchmark's different-entity tests are reached by arithmetic and the other's, mostly, are not.
4. A searched negative on magnitude: seven candidate predictors eliminated against a pre-registered floor with Holm correction, the best pairing of the survivors bounded in closed form at 0.534.
5. Every claim this project withdrew, kept on the page with what changed it - twelve claims, nine withdrawn or narrowed.

2. Related work

The nearest neighbour to §5.6, named first. Dong et al. (2025) identifies a flaw in RippleEdits' compositional evaluation and it is **not the one we measure**. Theirs is that the questions are built from an external knowledge graph, so *"they requires external knowledge beyond the edited content to correctly answer, thereby obscuring the assessment of logical generalization"* — a method is penalised for disagreeing with a KG about the intermediate fact. **Ours is about what the prompt contains rather than what the question requires:** the compositional prompt names the edited subject verbatim, so the update arrives at that token by arithmetic, whatever the model believes in between.

Their fix is the clearest illustration we have of why that distinction matters. They propose three repaired variants, and none removes the edited subject — the third, *In-Prompt*, *"explicitly inject[s] intermediate knowledge into prompts"*, making each question **longer** while the subject stays exactly where it was. A benchmark deliberately repaired by authors who had found a real evaluation flaw would still fail the screen in §8.1, because augmenting context does not move the token an update is delivered to.Read in full rather than from the abstract: the body contains no instance of *surface*, *lexical*, *overlap* or *verbatim*, and *subject* appears only in template grammar and path endpoints. R-018

We do not claim the surface-form result. Section 3.1 depends on the fact that scoring answers by string probability is distorted by how the answer is spelled. That is surface form competition Holtzman et al. (2021), and the correction we independently rebuilt — score against a subject-free version of the same prompt — is essentially their Domain Conditional PMI. We rediscovered a 2021 result and its standard remedy, and cite it to disclaim novelty.Found by a prior-art gate run *after* the measurement and *before* publication, which is later than it should have been.R-010

We do not claim knowledge-status × editability. Whether a model already knowing a fact predicts how well that fact edits has been measured, on Llama-3.1-8B, with MEMIT, AlphaEdit and fine-tuning (Pan et al., 2025). That closed a direction we had designed and were about to build.Two further gates cleared rather than closed: R-002 on intra-memory conflict and uncertainty-based deferral, and R-007 on the nearest remaining competitor.

Template sets exist upstream. CounterFact records derive from ParaRel entries containing hand-curated template *sets*, and keep one prompt per record (Elazar et al., 2021); our probes inherit that collapse.

Backward probing is narrower than we first

we ask	true answer	log P
`By profession, Maurice de Vlaminck is a`	painter	**−0.38**
`Maurice de Vlaminck is a citizen of`	France	**−1.12**
`The native language of Maurice de Vlaminck is`	French	**−0.50**
`Maurice de Vlaminck was born in the city of`	Paris	*(generates* `" Paris, France,"` *)*

we ask	true answer	before	after	moved
profession	painter	−0.38	−5.56	5.19
citizenship	France	−1.12	**−6.50**	**5.38**
native language	French	−0.50	−3.38	2.88

claimed. RippleEdits' Logical Generalization covers inverse and symmetric relations, which is the closest existing thing to a grounds probe; reading its test-type definitions directly is what forced the narrowing.R-001

Neither the depth result nor the layer sweep is ours. He et al. (2026) edit GPT-J-6B with ROME across layers 5–20 and report paraphrase generalization falling from 92.5% to 31.7% on MQuAKE — §5.4's phenomenon, published January 2026. They also report *"the average cosine similarity between subject keys for original and rephrased prompts declines steadily from roughly 0.80 at layer 5 to about 0.50 at layer 25"*, which is §5.3's decay and the cosine decomposition beneath it. We measured 0.984 → 0.471; they measured 0.80 → 0.50. Same quantity, same shape, eight months earlier.

What survives is narrow, and sharper than "we had a control". Their warrant for reading the effect as depth is that *"ROME achieves stable and high edit success rates"* across layers. **Edit success is the metric §5.4 shows cannot measure edit strength** — lift on the injected target spans 0.17 nats across two layers and a fourfold step budget while belief displacement spans 5.70. So the inference rests on a check we independently measured to be uninformative, and matching on displacement instead (a 16× step budget) is what licenses the comparison. The contribution is not the phenomenon and not the mechanism: it is that **the field's standard efficacy check cannot support the inference drawn from it.** §8.1's fourth question asks whether that is worth anything.Both sentences above were written from this paper's abstract and corrected after reading it. An abstract under-reports overlap and over-reports applicability, because it sells the contribution and buries the setting.R-012

What remains unclaimed by them: that prefix-sharing probes sit at **exactly 1.000**, for any `C`, at every layer — and §5.3's finding that the norm ratio carries none of the decay.Found by re-running prior art against where the project is now rather than where it started — after §5.4 was written. The paper is still unread; this is from its abstract, which is enough to establish the overlap and not to size it.R-011

A theoretical account that our measurements contradict. Ravfogel et al. (2026) prove, in a single-layer setting, that subject embeddings encode linear superpositions of their attribute vectors and the MLP acts as a relation-conditioned selector — explicitly *not* the associative key-value store ROME assumes. An earlier draft of this section offered it as §8.2's first candidate explanation for type-matched displacement. **Reading the paper withdraws that.** Their own limitations name *"typed attributes"* as outside the setting: attributes are arbitrary entities with no type structure, so the construction cannot predict a result that is entirely about type.

The disagreement runs deeper than absence. Their subject embedding is a **block per relation**, and their counterfactual perturbation experiment finds the MLP *"follows"* on the queried relation and stays *"stable"* on the others — perturbation local to the edited relation. §5.5 measures the opposite: damage to everything about the subject, worst where the *injected value's type* matches, not where the edited relation does. Either the single-layer construction does not carry to deep models, which they flag themselves, or ROME's rank-one update does not act on the structure it describes. **§8.2's first thread still has no candidate explanation.**

What is left. None of the above examines what an edit does to a fact's *premises*, and none reports the coefficient structure of the update itself. ROME (Meng et al., 2022) derives the $\mathtt{C}^{-1}$ term and its own repository enables it by default; EasyEdit disables it in all fourteen of its ROME configs, including for the two models the paper used.Read from both repositories rather than from either paper.R-009 Both values read from the raw config files.

3. Methods

Records: · · · E-013 gate

Possession gate. A chain is usable only if the model holds all three facts.The gate exists because CounterFact has none: its construction never checks the model knows the fact being edited, verified from

we ask	after a **Chile** edit	after an **urban planner** edit
profession	−5.56	**−9.59**
citizenship	**−6.50**	−3.06
native language	−3.38	**−0.62**

source.D-002 · R-006 The measure was built over E-004 → E-005, and E-006 established ROME on Llama-3.1-8B via NDIF is feasible at all. Each is scored against a type-matched candidate pool under two criteria: the true answer ranks first with the real subject (*row test*), and outscores it under foil prompts whose true answer differs (*column test*, an AUROC; chance 0.5). The column test replaces a placeholder control that is undefined for high-prior answers — see §3.1. 136 chains mined → 78 usable → 42 carried through the edit.The relation inventory behind the chains is R-003; the entailment structure and why grounds are evidential rather than deductive is E-001 · . Set-level coordination lift, which does not survive per-candidate, is T-055.The 29 missing chains were lost to backend outages, not excluded by any criterion — see . Every rate here is an existence claim, never a frequency.

Model choice. Possession is a *construct requirement*, not ambition: a model that never held the grounds cannot orphan them, which replaced this project's original size ceiling with a size floor. Possession rises with scale and the field's standard test overstates it at every sizeE-003; GPT-J-6B sits at 73%E-003b, and the 70B gate clears while showing the three chain legs are not independent — which is why §3's gate scores all three rather than the edited fact alone.

Edit. ROME as configured by EasyEdit, `C = I`, layer 5, `v*` optimised by gradient descent through the frozen model with EasyEdit's `llama3-8b.yaml` hyperparameters. Applied as a trace-time intervention computed from each position's own key, so the edited weight matrix is never formed — exact, not approximate.

Controls. Three, all required. A **same-subject** control edits an unrelated property (occupation) of the same person at matched magnitude, isolating content from subject-key perturbation. A **relation** control edits the same subject to the same target country through a relation that licenses nothing (*works in*). A **placebo country** gives the null rate for "lands in country X".

Readout. Full city names ranked over a 75-city pool of chain cities plus every target country's capital — widened deliberately.A pool that cannot express the coherent answer scores a correct relocation as a failure. That mistake is , and widening the pool is how it was avoided here.

3.1 Repairing the measure, which took two orthogonal fixes

Records: · ·

Before any edit: the possession measure failed on the most common answers. Rendering countries as Wikidata labels (`United States`) rather than naturally (`the United States`) is ungrammatical after *"born in the country of"*, and rank-1 on those answers sat at 21%. Rendering naturally moved it to **69%** — but then the placeholder control ranked the same answer first at the identical 69%, because *"[X] was born in the country of the United States"* is the modal completion whether or not `[X]` means anything.

Neither repair alone recovers anything. Together they recover twenty chains.

Table 1. Usable chains out of 136, by rendering × control. Each fix alone moves the

count by one; together they move it by twenty.

This is an interaction, not two additive fixes, and it is why each looked like a null when tested alone. Llama-3.1-70B under the old measure gave 74 usable chains; Llama-3.1-8B under the repaired one gives 78. Repairing the instrument was worth more than an order of magnitude of scale.

4. Implementation

This project built an instrument rather than a method, and the instrument is the record. There is no package to install: the measure is three modules and the thing that makes it trustworthy is that every run is written down.

Every record behind this paper is listed in the **evidence ledger** — all 56 decisions, 33 findings and 85 threads, generated from the repository so it cannot drift from it. Most of them went nowhere; that is the point of publishing the whole list rather than the ten claims this paper defends.

Whether a reading can be taken twice is answered in the sibling project rather than here, because it was found there: readings off a shared inference fabric vary by which replica served the request, exactly and not by drift, so a ratio read inside one trace cancels it and a ratio read against a cached value does not. See edit-reach §4.1.

5. Results

5.1 The update coefficient is pinned at the subject

Records: E-016 gate · E-016 result

ROME's update divides by `u.k*`. That one term fixes the coefficient across a whole class of prompts, and the class is larger than it looks.

`k*` is read at the subject's last token of the edit prompt. A probe beginning with the same subject has an identical key there, so the coefficient is `(k*.u)/(u.k*) = 1` for any `u`. The measured value is **1.0000**, not approximately one.

	placeholder control	paired control
bare rendering	58	59
natural rendering	59	**78**

evidence ledger	the complete record, generated
`agents/shared/decisions.md`	56 entries; falsification stated before each run
`agents/shared/findings.md`	33 literature entries, including the two that killed directions
`threads.md`	85 threads — 66 answered, 9 parked, 6 open, 3 dropped
`results/*.json`	per-item records, … ,
`logs/`	every run, levelled and committed, including the failures
`src/possession.py`, `src/discrimination.py`	the shipped measure and the paired control

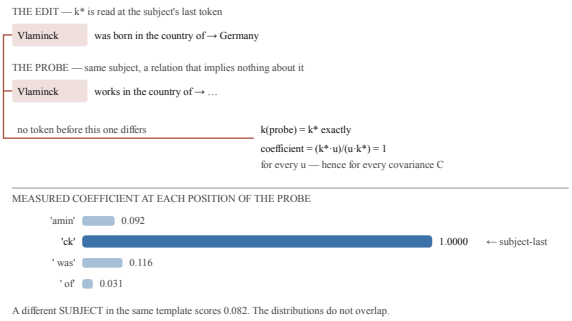


Figure 1: Why the coefficient is pinned. The two prompts share every token up to the subject's last. Causal attention makes the key there identical, and ROME's own normalisation divides by u · k*, so the coefficient is exactly 1 for any u.

Everything in §5.2 and §5.3 follows from this. The measurements came first chronologically — the narrative version keeps that order — but the dependency runs the other way:

Table 3. Every observation in the arc, against the one mechanism.

Table 4. A position ablation inside the update. Only the mask varies — same edit, same cached v, *same k*. Arm E restores the magnitude the mask removed, so position is the only difference from the full edit.

A coefficient of 1 at a position contributing nothing would produce the same numbers as one that does the work, so the arithmetic on its own does not settle where the effect lives. Masking the update by position does.

Sufficient: arm B reproduces the full edit from one position out of roughly twelve — 69% against 67%, one discordant pair, p = 1.000.

Necessary: arm C also strips two thirds of the magnitude, and §5.1's scale test showed 0.27× breaks relocation unaided, so C ≈ D is what magnitude loss alone predicts. Arm E restores that magnitude to the surviving positions and reaches 12% against a 5% baseline.

The comparison that carries it is **B against E**. Arm B has *less* mass at one position; arm E has *more*, spread over every other position. **29 against 5, twenty-four discordant pairs, none reversed, $p < 0.0001$.** Coefficient mass at the subject token ranges 31–80% across chains, mean 66.8%. Arm E's rescaling is per chain, so each restoration is exact rather than averaged.

Less magnitude in the right place beats more magnitude everywhere else, and no chain goes the other way.

5.2 What it predicts: displacement without inference

Records: · ·

§5.1 predicts that a probe beginning with the edited subject receives the full update whatever it asks about — so an edit should move a *premise* as readily as a consequence, and a semantically unrelated edit should move it just as far. Both hold.

The birth-city probe lands in the edited country 67% of the time (baseline 5%, same-subject control 5%, placebo 0%). Edinburgh → Hamburg for Germany; Paris → Santiago for Chile.

That 67% is an undercount, and the cause is ours. Letting the edited model generate freely instead of ranking the pool, it names a city in the target country in **30 of 30** chains. The 75-city pool cannot express most of its answers — Hamburg, Sydney, Zurich, Milan, Gisborne and Plzeň are all absent — so ranking falls back to the highest-prior item on the menu.Which is also what the `Washington, D.C.` attractor was: 33% of ranked destinations, read at the time as the model's fallback. It was the ranking's. **Differences survive a bad pool; absolute levels do not** — §5.2 and §5.3 are paired contrasts using the same pool in every arm, so the artifact cancels there and only this level is affected.

Then the relation control. The two edits differ in one word, and only one of them licenses any inference about a birth city:

```
arm A   Maurice de Vlaminck was born in the country
↪  of   -> Chile
arm B   Maurice de Vlaminck works in the country of
↪     -> Chile
```

Both are then asked `Maurice de Vlaminck was born in the city of`:

Table 2. The relation control. Subject and target country held fixed; only the

observation	explanation
birth-city probe displaced, 67%	shares the prefix → pinned at 1.0
"Paris is located in…" inert, −0.02	shares no prefix → nothing pinned
whitening changed nothing, 42/42	cannot alter a coefficient fixed by arithmetic
a uniform ×0.27 rescale **does** break it (3 of 4)	it scales the pinned term too
the work-country null, +0.0 pp	both prompts begin with the subject

arm	positions receiving δ	mass	in target
A full	all	100%	28/42 = **67%**
B only	the subject's last token alone	67%	29/42 = **69%**
C except	all but it	33%	2/42 = 5%
E except, **mass restored**	all but it, rescaled to 100%	100%	5/42 = **12%**
D none	—	0%	2/42 = 5%

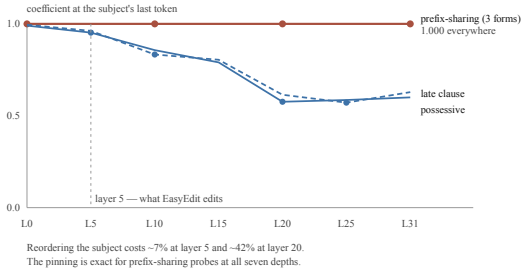


Figure 2: Coefficient at the subject's last token, by layer, 12 subjects. Prompts beginning with the subject are pinned at exactly 1.000 at every depth. Reordering the subject escapes progressively — but not at layer 5, which is the layer this configuration edits.

edited relation varies. Exact McNemar p = 1.000, nine discordant pairs each way.

Exact McNemar p = 1.000 (nine discordant each way). Where both arms land in the target country, they choose the *same city* 84% of the time. Editing where a person works relocates their birthplace exactly as often as editing where they were born. ### 5.3 Scope: the claim's boundary, measured

Records: · · · · ·

The five forms, as fed to the model — the first three begin with the subject, the last two do not:

```
edit form            Maurice de Vlaminck was born in
↪  the country of
same-prefix probe    Maurice de Vlaminck was born in
↪  the city of
different relation   Maurice de Vlaminck died in the
↪  city of
late clause          The city where Maurice de
↪  Vlaminck was born is
possessive           The birthplace of Maurice de
↪  Vlaminck is the city of
```

A separate control varied the **subject** while holding the form fixed: chain i's `k*` scored against chain $i{+}1$'s subject in the same template. Mean **0.082**, max **0.153** — below the *minimum* same-subject value of 0.483. The distributions do not overlap, and the prestated confirmation threshold was ≤0.3.

The decay is a change of direction, not of magnitude. The coefficient is `(||k||/||k*||).cos(k, k*)`, so it can fall two ways, and they are different claims about the network. Across eight layers, six forms and 16 chains, the cosine falls **0.984 → 0.471** while the norm ratio never leaves 0.97–1.24 — and drifts slightly *up*, so magnitude works against the decay. Subject-position keys have essentially the same length regardless of whose name is in the prompt or how it is worded; everything that distinguishes them is orientation.This does **not** show that attention mixes context into the subject position — a falling cosine is equally consistent with MLP writes there. *Measured since:* decomposing the divergence into per-layer attention and MLP writes, **attention never leads anywhere** — across three prompt forms and eight layers its largest share is a ratio of **1.03**, while the MLP leads about 2× early and 3× at layer 31. The verdict is still BOTH (0.81, inside a pre-stated [0.5, 2.0]), so the account is not settled; but the candidate named here is directionally contradicted at every depth. · The different-subject floor's rise at layer 31 (0.082 → 0.669) is likewise cosine with a flat ratio: representational convergence near the output, not a scaling artifact.

It is a property of ROME, not of this model, and not of this scale. The derivation mentions neither architecture nor parameter count, so it should hold anywhere. Measuring rather than assuming it costs almost nothing, because the coefficient needs **no gradients** — one forward pass, save the MLP input at the subject's token.

Table 9. Prefix-sharing probes across a 68× parameter range and two

architectures. Every cell is exactly 1.000; the column records the worst deviation found.

The **decay** is approximately scale-invariant at matched depth — worst deviations 0.131 and 0.144 against a pre-stated 0.15 band, which is a pass close enough to the threshold to be worth saying so.The 405B arm is instruction-tuned, because it is the only 405B hosted. It varies scale *and* post-training together, and post-training is exactly the sort of thing

arm	lands in target country
`X was born in the country of` → T — licenses the inference	28/42 = **67%**
`X works in the country of` → T — licenses nothing	28/42 = **67%**
paired difference	**+0.0 pp**

model	params	layers	prefix-sharing probes
GPT-J-6B	6B	28	**1.000** · worst 0.0e+00
Llama-3.1-8B	8B	32	**1.000**
Llama-3.1-70B	70B	80	**1.000** · worst 0.0e+00
Llama-3.1-405B-Instruct	405B	126	**1.000** · worst 0.0e+00

that could move subject representations. It must not be cited for scale alone.

The *decay* transfers too, which was not expected, because it is not derived. Compared at matched fraction of depth both models fall from 1.0 at the input to ~0.5 by two thirds depth and flatten — Llama 0.951 / 0.577 / 0.598 against GPT-J 0.861 / 0.597 / 0.482 at depth fractions 0.15, 0.64 and 1.00.

The floor does not transfer, and that narrows §5.3. GPT-J's different-subject floor is **0.155** at the edit layer against Llama's 0.082, and **0.524** at the final layer — above that layer's same-subject late-clause value of 0.482, so the coefficient has stopped telling subjects apart at all. *"Different subjects are near-orthogonal at the subject position"* is a **Llama-3.1-8B** statement, not a ROME one. The margin at the layer ROME actually edits stays clean on both (0.87 against 0.71), so the claim above is untouched; only its generality across depth is model-dependent.

Floor magnitude is model-specific and is not a scale law. Mean floor at depth fractions 0.10–0.23: GPT-J **0.144**, Llama-8B **0.061**, Llama-70B **0.110**, Llama-405B-Instruct **0.012**. The 405B figure is remarkable — 0.009 at fraction 0.10, with a maximum across all twelve subjects of 0.042, against a same-subject value of 0.992, a **108× contrast**. But 70B is *worse* than 8B, so the trend is non-monotone on four points and no scale law follows from it.Two structures do hold in all four: the floor is high at layer 0, before contextualisation, when the key is embedding-driven and names share subword statistics; and near the output the floor *exceeds* the same-subject value, so the measure stops discriminating subjects altogether.

What the floor does across depth, in every model measured: it is lowest in the first fifth of the network and rises monotonically after — a **14× range** on Llama-3.1-8B, 0.048 to 0.697. An earlier version of this section reported the minimum as falling near the layer ROME edits, in two models, and drew a line to ROME's layer selection. Measured at finer spacing that is **false**: the minima sit at fractions 0.06 and 0.11, not the same place, and Llama's edit layer at fraction 0.16 carries a floor of 0.095 — about twice its minimum.Withdrawn rather than qualified. The claim was attractive, and it did not survive its own first proper test.Tested rather than assumed. Regating the subjects on GPT-J and splitting them gives a floor of **0.146 for held and 0.146 for not-held** — a gap of 0.000 against a pre-stated band of 0.05, with every gap running the *wrong* way for a familiarity account, and GPT-J holding 21 of 36 subjects outright. The narrowing is confirmed. The residual objection is that a candidate pool undercounts, which dilutes the contrast toward exactly this zero. ·

5.4 Delivery is not effect — the coefficient's boundary, measured

Records: · ·

Everything above measures **delivery**: how much of the update vector arrives at a position. §7.3 listed the obvious objection — delivery is not behaviour — as one we could not answer. We can now, and **the objection is correct**. The thread is ; the run that asked first and got it wrong is .

Apply the update at layer 20 instead of layer 5. A subject-initial probe receives a coefficient of **exactly 1.000 at both**, analytically, since both share the edit prompt's prefix up to the subject's last token. Delivery is therefore held fixed by construction, and anything that differs is not delivery.

Table 6. Propagation to a prefix-sharing probe at a delivered coefficient of

exactly 1.000 in every cell. Displacement is the drop in log P(true answer) on the edit prompt itself — the edit's strength, measured where it was optimised.

> **1.22 nats of reach per nat displaced at layer 5, against 0.41 at layer 20 — a factor of 2.96, holding across a sixteenfold range of optimisation budget.**

Two confounds had to die first, and the order matters because our first attempt got it wrong. A layer-20 edit under EasyEdit's shipped hyperparameters displaces only 2.61 nats against layer 5's 7.26, so the first run of this comparison was measuring a **weaker edit**, and its conclusion — this conclusion — was withdrawn.The gate that let it through measured *lift on the injected target*, which a logit nudge also achieves. Lift spans 0.17 nats across both layers and a fourfold step budget while displacement spans 5.70: it saturates by 25 steps and carries no information about

cell	displacement	propagation	reach per nat displaced
layer 5, 25 steps — EasyEdit's shipped config	7.26	9.66	**1.33**
layer 5, 400 steps	11.74	12.96	**1.10**
layer 20, 25 steps	2.61	0.80	**0.31**
layer 20, 400 steps — displacement-matched	6.97	3.59	**0.52**

edit strength thereafter. Matching took a **16× step budget** at layer 20. That match then made *layer* collinear with *optimisation budget*, so a fourth cell was bought to separate them: more optimisation buys **more** reach — +34% at layer 5, +349% at layer 20 — which is the opposite sign to the overfitting account.

What this costs the sections above. They describe delivery, and must be read that way. The claim is unchanged in content and narrower in what it licenses:

> Any probe beginning with the edited subject receives the full, unattenuated edit vector at the subject position — at every layer, for any `C`, regardless of relation.

What no longer follows is *how far that probe's answer will move*. §5.1's position ablation stands as a **layer-5** result.

5.5 An edit displaces probes whose answer type matches the injected value

Records: · · ·

If the unit an edit operates on is the **subject**, the relation it targeted is incidental — so a birthplace edit should disturb attributes of that person having nothing to do with birth. It does, and not uniformly.

Three probes per subject, teacher-forced log P(true answer), paired against the same probes on an unedited subject, no candidate pool. The probes, literally, with Vlaminck substituted in:

```
occupation        By profession, Maurice de Vlaminck
↪  is a      -> painter
citizenship       Maurice de Vlaminck is a citizen of
↪           -> France
native language   The native language of Maurice de
↪  Vlaminck is -> French
```

and the edit that precedes them is `Maurice de Vlaminck was born in the country of` → `Chile`. Averaged over 29 subjects:

Table 7. Post-edit level of each attribute after a birthplace edit. Levels,

not drops — the probes start far apart, so drops are baseline-dominated and reading them as targeting is how this run's first conclusion went wrong.

Occupation and language finish **0.04 nats apart after starting 3.65 apart**. Citizenship finishes **2.95 below** both. Semantic relatedness predicts language patterns with citizenship; it patterns with occupation instead. The edit is not reasoning about origin, it is competing for a slot.

That reading proposed **two** effects — a floor, and type-matched displacement. Editing a *different relation* tests both, and they do not survive together.

Table 8. The crossover. Inject a profession instead of a country and the

displaced probe should swap. Six arms: two models, two relations, two depths.

Type-matched displacement: 6 of 6. Swap the injected value from a country to a profession and the hardest-hit probe swaps with it — occupation's displacement more than triples while citizenship's roughly halves. Difference-in-differences **+8.06 nats** at 8B and **+5.42** at 70B layer 13, matched by depth fraction rather than layer index.

The floor: 1 of 6, and it is withdrawn as a general claim. Only the 8B birthplace cell shows the convergence that named it. That cell is not a fluke — a fresh optimisation reproduced it at **0.09** against the **0.04** above — so the floor is real, replicable, and **specific to one model-relation pair**. It is not a property of editing, and the "two effects" sentence above is superseded by this one:Recorded rather than silently revised: §5.5 asserted a general floor for nine days on the strength of a single relation. The crossover was the first design that could have refuted it, and did.

> An edit displaces probes whose **answer type** matches the injected value. Whether everything else converges to a common level is a fact about the particular model and relation, not about the method.This run's own headline said "semantic relatedness", computed on drops, and was retracted. The confound was named in the ticket that commissioned it while the code still computed on drops — naming a confound is not controlling for it.

What predicts how far an edit travels is mostly unmeasured, and we have now searched. Belief displacement and ROME's efficacy margin each correlate with propagation at Spearman **0.42** — about 18% of variance — and they are not two views but one: ROME optimises `v*` until the target saturates, so post-edit log P(target) has a standard deviation of **0.02** against the true answer's **5.68**, which makes the efficacy *margin* rank-identical to the true answer's suppression at $\rho = \mathbf{1.000}$ **exactly**.A margin that appears to compare two quantities compares one. This also sharpens what the margin can be

probe	type-matched to the injected value?	related to birth?	post-edit level
occupation	no (a profession)	no	**−6.16**
native language	no (a language)	yes	**−6.20**
citizenship	**yes** (a country)	yes	**−9.10**

model	layer	depth frac	edited relation	type-matched probe moved most?	non-matched spread
70B	5	0.06	birthplace	**yes** · citizenship	1.56
70B	13	0.16	birthplace	**yes** · citizenship	1.11
70B	5	0.06	occupation	**yes** · occupation	1.43
70B	13	0.16	occupation	**yes** · occupation	1.02
8B	5	0.16	birthplace	**yes** · citizenship	**0.09**
8B	5	0.16	occupation	**yes** · occupation	2.73

used for: it is a suppression measure wearing a comparison's name.R-015 The coefficient cannot close the gap either — pinned at exactly 1.000 across precisely the probes whose outcomes differ most, it has **zero variance where the outcome varies**.

Seven candidates were then eliminated against a pre-registered floor of $\rho = 0.50$ with Holm correction: update size, key size, their ratio, the belief's strength, the target's prior, the subject's length, and the local residual norm. None cleared.Two of the seven cleared an *unadjusted* α and neither cleared its Holm threshold — the correction changed the answer rather than decorating it. The strongest, `||Δv||` at **−0.423**, is reliable after correction and sits below the effect floor, and its sign is backwards: **a bigger update travels less far**, against the rationale it was listed with. The best *combination* of the two surviving predictors has a closed-form ceiling of **0.534** — fit on the data it is scored against, so maximally optimistic — which fails a 0.60 bar by 0.066 before shrinkage and by 0.101 after.That ceiling was computable from three numbers, and it replaced a planned run of 94 fresh edits. R-016

So nothing explains more than about 18% alone or 28% together, and that now rests on an exhaustive search of the candidates this record contains rather than on not having looked.

5.6 A benchmark's different-entity tests still name the edited subject

Everything above is measured on probes we wrote, and the obvious objection is that our probe shapes produced the result. RippleEdits (Cohen et al., 2024) ships 4,755 edits and 46,754 test prompts sorted into six criteria, none of which we designed. So we ran the identical measurement on **their** edits and **their** prompts, rewriting nothing: `k*` read at the subject's last token of each entry's own edit prompt, the coefficient read at the subject's last token of each test prompt, layer 5 — the layer EasyEdit's shipped `llama3-8b.yaml` edits.

Table 11. Fraction of a ROME update arriving at RippleEdits' own test prompts, against a floor built from the same prompts. Llama-3.1-8B, layer 5, 400 entries of the recent subset, 193 skipped for want of a recoverable subject label.

The second column is the whole result. A near-1 coefficient means nothing without a floor, and importing the **0.082** of §5.3 would be an assumption rather than a control: that number came from our probe set in a different prompt register. So every test prompt was scored a *second* time against a **different entry's** `k*` — partner drawn by a seeded derangement that rejects any partner sharing a token with the entry's own subject. Same prompt, same layer, same token index, same arithmetic; only the edit changes. The floor is **0.070**, tight enough that no criterion's quartile touches 0.16.The failure this was built to catch: RippleEdits' prompts are templated, so if a template's keys overlap heavily regardless of subject, the coefficient would sit near 1 for any pairing and the whole table would be a fact about the measure. Pre-stated as the outcome that would withdraw it. It did not occur.

The row to read is Compositionality I, at n = 1,981. Their compositional tests probe a fact about a *different* entity, reached through the edited one — and they sit at **0.982** against a floor of 0.065. The reason is visible in the prompts: *"the name of the head of government of the place of death of* ***Leo Arons*** *is"* **names the edited subject verbatim**. A test designed to ask about a different entity still puts the edited subject in the prompt, and at that token it receives essentially the whole update.We predicted a floor here and were wrong, then misdiagnosed our

criterion	what it varies	n	own k*	IQR	permuted k*	IQR
Forgetfulness	same entity, negated relation	241	**1.000**	0.000	0.072	0.047
Compositionality I	a different entity, one hop	1,981	**0.982**	0.028	0.065	0.045
Logical Generalization	same entity, inverse/symmetric	208	**0.970**	0.058	0.069	0.037
Relation Specificity	same entity, different relation	1,818	**0.969**	0.046	0.075	0.048
Subject Aliasing	same entity, different string	27	0.898	**0.468**	0.071	0.044

own error as a measurement artifact before checking their prompt text. The correction runs against us: the measurement was right and the prediction was wrong.

What this does not establish. Subject Aliasing is the one criterion that could separate a key on the *entity* from a key on the *string*, and it is the one we cannot report. Aliases with a genuinely different surface form sit at **0.302** against **1.000** for prompts that merely contain the canonical label — the 0.468 IQR above is those two populations in one row — but at n = 9, below a threshold of 30 fixed before the run. The gap looks decisive and is not reported as one.Since measured at n=370 over all three subsets: aliases sit below same-entity in every one, and the drop is **graded by how far the string moved** — $\rho = +0.778$, slope +0.847, so delivery to a renamed subject is roughly its string similarity to the original. The subsets' apparent disagreement is composition, not a popularity effect: controlling for similarity, a Wikidata-sitelink popularity proxy predicts nothing (partial $\rho = +0.043$). · · An earlier run concluded the opposite from eight near-identical strings, its entry filter having required the canonical label to appear — which is exactly what an alias does not do. That sentence was withdrawn, and the threshold exists because of it. And as everywhere above, this is **delivery**: §5.4 put delivery and effect 2.96× apart, and nothing here measures how far an answer moves.

Is that construction universal, or RippleEdits'? We counted, because the answer decides whether the paragraph above is a finding or a note about one dataset. Over all four released MQuAKE (Zhong et al., 2023) files — 51,258 multi-hop questions — the edited subject appears in **57.9%**. The aggregate is noise. Restricting to single-edit instances, where which hop carries the edit is unambiguous, the structure underneath it is exact:

Table 12. MQuAKE multi-hop questions containing the edited subject's string, by which hop of the chain carries the edit. Single-edit instances; four released files, fetched 2026-10-14.

Not approximately — across 23,367 questions, with no exception in either direction. The reason is structural: MQuAKE phrases a chain from its head, so an edit on hop 2 is an edit to an *intermediate* entity the question never mentions. "*What is the capital city of the country that **Michael Feinstein** is a citizen of?*" — the edit is on *United States of America → capital*, and the question names Feinstein instead. MQuAKE-T's 0.4% is the same fact: 1,861 of its 1,868 single-edit instances edit hop 2 or later.R-014

So neither reading we expected is right, and the real one is more useful than either. RippleEdits' compositional tests always name the edited subject because its edits are always on the head entity — Table 11's 0.982 is not a quirk of our reading. MQuAKE's hop-2-and-later questions are the genuine different-entity test the field is assumed to have, and there are 16,506 of them. **One benchmark contains both populations, separated exactly by a property recorded in its own metadata.** That makes MQuAKE a within-benchmark control nobody has to construct: hop-1 against hop-2+ questions, one dataset, one prompt register, one set of edits, with the cross-benchmark confound gone by construction. It is the measurement we would run next.

We ran it, and it put a caveat on this whole section that is worth more than its own result. Measured on MQuAKE's two-hop chains at the same layer: a question that names the edited subject receives **0.933** at that subject's token, and one that does not receives **0.055** — the floor — at every position it has. The structure holds. But the arm we had pre-registered as the headline, reading both at the **final** token, returned 0.027 against 0.055: *neither* arm receives anything where the answer is produced.A design error worth recording: the final token was chosen because it is the only reading immune to a positional artifact, which is true and incomplete. Nothing in our own lenses asked whether the chosen measure *varies* before it was fixed, and a comparison with no variance cannot discriminate.

Within a single prompt, then, the subject token and the final token are a factor of **35** apart. So every

hop carrying the edit	questions	names the edited subject
1	6,861	**6,861 — 100.0%**
2	13,977	**0 — 0.0%**
3	2,313	**0 — 0.0%**
4	216	**0 — 0.0%**

number in Table 11 — the 0.982 included — is a property of **a token, not of a prompt**, and the answer position is not that token.

This does not say the edit fails to reach the answer, and the table must not be read that way. The coefficient measures *direct* delivery at layer 5; an altered subject representation still reaches the final position through attention at layers 6 and above, by a path this measure cannot see. What it does say is that "this prompt receives 98% of the update" is shorthand for "one token in it does" — which is §5.4's delivery-versus-effect gap in its positional form, and the sharper statement of it.

One thing we did not design for: the floor has now been built two independent ways — a permuted `k*` on RippleEdits (**0.069**) and a different token on MQuAKE (**0.027**–**0.055**). Different corpus, different construction, same range.

6. Findings and claims

Every row below states the observation that would change its status. For the one claim still standing that is what would refute it; for a claim already withdrawn or narrowed it is what would bring it back. Both are the same kind of sentence, and a row that cannot carry one is not a claim — it is an opinion with a citation attached.

Twelve claims. **One stands, two are narrowed, nine are withdrawn** — and the one still standing is analytic, which is to say no measurement was ever going to take it down.This sentence used to read "nine withdrawn or narrowed", which takes the count of the withdrawals and then describes it as covering the narrowings too. Eleven of twelve did not survive intact. Corrected 2026-11-09. That ratio is the honest summary.

The last row is one we never claimed and checked anyway. §7.3 carried it as an objection we could not answer; answering it took four runs and cost us the strongest available reading of our own mechanism.

6.1 What the withdrawals rule out, and the boundary they leave

Table 5 is a list. This is what it adds up to — and the first thing to separate is **elimination** from **our own cost**, because only one of them is a contribution and a record that mixes them earns credit for neither.

Eliminations. These constrain the space for anyone working in it.

Our own cost. These constrain nothing and are not offered as findings.

A gate that measured target lift, which a logit nudge also passes (). A dip criterion defeated by a flat-bottomed minimum (). Layer 5 used on two models where it is depth fraction 0.16 and 0.06 (). A delta cache that reported a partial run as a whole one. They are in the record because they are why the rest can be trusted — not because they narrowed anything.

The boundary the eliminations leave. Stated as sharply as the evidence allows:

> **Achievable.** *Whether* an edit reaches a given prompt — exactly, with no forward pass through the edited model, at any scale from 6B to 405B. And *which* probes take the worst of it — the injected value's answer type predicts it, 6 of 6 arms.
>
> **Not achievable, on this record.** *How much* any of them moves. Belief displacement and the efficacy margin reach $\rho \approx 0.42$ — about 18% of variance, and they are one quantity, not two — while the coefficient cannot help by construction, pinned at exactly 1.000 across precisely the probes whose outcomes differ most.

We can say which facts an edit touches. How hard is boundable but not predictable. That sentence is the whole contribution of the negative results, and it is now a searched claim rather than an unexamined one: seven candidates eliminated against a pre-registered floor, and their best pairing bounded at **0.534** in closed form.Worth saying plainly because the two read very differently to a reviewer: "we found no predictor" and "we tested these seven and bounded their combination" are not the same sentence. · R-016

This paragraph read "We cannot say how hard" until 9 November 2026. Two measurements in `edit-reach` changed it: an upper bound that exists and forbids nothing, and a second class of candidate eliminated. The claim is narrower now and says more.**Magnitude is bounded above, uninformatively.** Because the update's direction is fixed, Cauchy–Schwarz bounds the delivered fraction for every prompt in one line — so an upper bound does exist, which this page previously implied was unavailable. It is useless where it would matter: its median far-field value permits a question that never names the edited subject to receive *more* of the update than the near field actually receives.

What *does* set the magnitude is , and it still has no candidate. **Two classes of variable have now been searched — the geometry of the key, and the identity of the subject — and neither car-**

ries it. The second came back null across eight relationships, every interval spanning zero, over 41 subjects; what that rules out is a monotone relationship of roughly $|\rho| > 0.37$ and nothing smaller. The absence is evidence of having looked, and it is now specific enough that a reader can tell what has *not* been looked at.

7. Limitations

1 · n = 42, one model, one relation family, one layer. Every statement here is an *existence* claim; no rate is defensible and the artifacts are written to avoid implying one.A standing constraint rather than a disclaimer added at write-up: hand-built and mined sets license existence and never frequency. If the mechanism needs a frequency result to matter, that objection stands and we cannot meet it.

2 · The mechanism is a line of algebra — is it interesting? A reviewer could reasonably say practitioners already assume this. We have not found it stated, and "we did not find it" is the weakest form of novelty claim.

3 · Coefficient is not behaviour — conceded, and now measured. This was listed here as an objection we could not answer. §5.4 answers it against us: holding delivery fixed at a coefficient of exactly 1.000 and matching the edits on belief displacement, a layer-5 edit still reaches **2.96× further** than a layer-20 one. The coefficient **bounds delivery and does not predict effect**, so every section above describes delivery only. The original form of the objection was right on its own terms too — rescaling by 0.27 broke relocation in 3 of 4 chains while whitening's 3.7× attenuation changed nothing, so the map from coefficient to destination was never monotone.We asserted §5.4's conclusion once on a design whose arms differed in edit strength, and withdrew it. Earning the same sentence took three further runs: matched displacement, matched norm, and a budget control.

4 · `C = I` is what *EasyEdit users* run — and two implementations circulate. Verified from both repositories: `kmeng01/rome` sets `mom2_adjustment: true` for gpt2-xl and gpt-j-6B; EasyEdit sets it `false` for all fourteen configs including those two. So the whitening term the method is built around is on by default in the authors' code and off in the toolkit most third-party work reaches for. Our earlier phrasing — "what the field runs" — was too strong and is corrected here.This does not touch the central claim. measured the whitened arm too, and showed the mechanism lives in the `u.k*` normalisation, which both implementations share. Our whitened arm also estimated the covariance over the CounterFact prompt distribution rather than Wikipedia, from 2048 keys, with a ridge chosen by an in-sample/held-out agreement rule. Defensible, but not ROME.

5 · The causal claim is about the edit, not the model. §5.1's ablation intervenes on ROME's own update and shows which positions the relocation depends on. It does **not** patch the model's activations, so it says which part of *the edit* matters and never which part of *the network* represents the fact. Interchange interventions remain unrun, and the distinction between "this position of the update is necessary" and "this component encodes the fact" is exactly the gap that remains.

6 · Every absolute rate here is measured by ranking a closed pool, and that undercounts. Free generation relocates in 30 of 30 chains where ranking gives 67%. We report paired contrasts precisely because they survive this — but any single rate in this paper should be read as a floor, not an estimate.

7 · The stronger version of our own measurement critique is refuted — by us. A plausible reading of §3.1 is that possession is not a fact-level property at all, only a property of the (fact, template) pair, which would make every per-fact possession number here and in the literature a category error. We tested it against ParaRel's own P19 templates: the modal-cell share is **0.923** against a chance value of 0.125, and 65% of facts are identical across all eight templates.60 facts × 8 templates. All eight templates sit between 65% and 72% HELD, so no single bad probe carries the stability. Single-template probing is a noisy estimator of a real property, not a category error. The critique that survives is narrower: a cell can flip on phrasing, which biases a single-template estimate — not that there is nothing to estimate.

6 · Self-review has a ceiling. An adversary agent checks claims against this repo's own record and caught one overclaim before publication. It cannot catch an error the whole record shares — which is why this page exists.

8 · This is not a method, and the nearest misreading is a specific one. The pinned coefficient explains why an edit reaches every prompt sharing its subject. It is a consequence of how ROME normalises, not a technique on offer. The nearest misreading is *"edit a deeper layer to leak less"* — and as of §5.4 that is no longer blocked by a missing measurement, because we made it. A layer-20 edit does reach less far per nat of belief displaced, by a factor of 2.96.

We still do not offer it, and the reason is in our own numbers rather than in scope policy. Matching layer 5's displacement at layer 20 took a **16× step budget**. And what we measured falling is reach to *any* prefix-sharing probe — which includes the paraphrases an editor **wants** to reach. Our probes cannot separate leakage an editor would pay to remove from generalisation it would pay to keep, so "leaks less" and "generalises less" are the same measurement here. That is the field's specificity/generalisation tradeoff, and naming which side of it a deeper layer lands on is work we have not done.Reporting that a knob moves a quantity is measurement. Recommending the knob means showing

the trade is favourable, which would mean owning a baseline.

This entry exists because an adversarial framing has no natural slot for it. "What to attack" collects what a reviewer would dispute, and nobody disputes a constraint on use — so the category has to be added deliberately or it vanishes in the rename.

8. Conclusion

8.1 What we would like decided

0 · Is the screen worth having? Everything above is evidence for one small instrument, and this is the question we most want answered. A propagation test can only measure propagation if its test prompt's key does **not** overlap the edit's key — and `(k.k*)/(k*.k*)` decides that **before any edit is applied**, from one forward pass through the *unedited* model.Searched and not found, which is the weakest positive form. The nearest work inverts it: (Dong et al., 2025) and the subspace-confinement line treat key overlap as something an *editor* should manage, not as a property of an *evaluation item*. R-017

The rest of this paper is what makes the screen usable. §5.1 says it is **exact** rather than heuristic — analytic, four models, a 68× parameter range. §5.6 gives it a measured **floor** of 0.070 on a benchmark's own materials, so "near 1" and "near the floor" are calibrated rather than asserted. §5.4 says why it is **necessary and not sufficient**: a high coefficient does not tell you the answer moved, so delivery can never be substituted for effect. And §5.6's MQuAKE count is an existence proof that benchmarks satisfying it already exist — 16,506 prompts whose edited subject is absent, in a published artifact, by construction rather than design.

> **The question:** is a cheap pre-run criterion for whether a propagation item can test propagation something the field would use — or is it a restatement of what benchmark authors already do by eye?

1. **Is the pinned-coefficient result known?** The question we cannot answer from inside.
2. **Does it matter for ripple benchmarks?** *Rewritten 2026-10-14; the earlier version of this item asserted that every propagation benchmark probes with prompts beginning with the subject, and priced a week of work on it. We had inspected none.* Counting all three released subsets of RippleEdits (Cohen et al., 2024) — **46,754 test prompts** — **44 begin with their subject.** 0.09%. Of the prompts whose position can be determined, 99.8% put the subject *later*; nearly every one has the form *"The relation of subject is"*.Twenty minutes, not a week: the released prompts carry the subject as a Wikidata id and either materialise it or leave the slot empty, so its *position* is visible without resolving a single entity — and position is the whole claim.R-013
   The consequence survives, in a form we can now state against the benchmark's own materials. §5.3 measured subject-*late* probes at **0.951** and **0.962** at layer 5 — the layer EasyEdit's shipped config edits. RippleEdits' own prompt form therefore receives about **95%** of the update at the standard edit layer. The concern was never that a probe begins with the subject; that was our probes' shape, not the mechanism's requirement. The mechanism needs only a high coefficient, and at the layer the field edits, both forms have one.
   What the correction forecloses is the depth half: at layer 20 that form falls to ~0.58, so a deep edit would reach these prompts far less. We were not claiming that, and now cannot.
   That measurement has since been run, and §5.6 is it. On their edits and their prompts: **0.969–1.000** for the same-entity criteria, **0.982** for the compositional ones, against a floor of **0.070** built by scoring every prompt a second time against a stranger's `k*`. The compositional row is the one we did not expect — their tests about a *different* entity still name the edited subject verbatim, so the update arrives in full at a prompt designed to probe something else.
   What we would like decided is now narrower and sharper: is "a benchmark's different-entity tests receive the whole update at a token they did not intend to be load-bearing" a finding the field would act on — or the expected behaviour of a subject-keyed method, interesting to nobody who already knows ROME is subject-keyed? We cannot tell from inside, and it is the difference between a result and a restatement.
   And a second question we would rather have asked than answered. §5.6 ends by showing the coefficient is a property of a token rather than of a prompt, with the subject and final positions a factor of 35 apart in the same forward pass. Everything this paper measures is delivery at one token. Is a delivery measure that cannot see the attention path to the answer position worth building on — or is the honest next instrument one that measures arrival at the position the model actually reads?
3. **Is n = 42 with an existence claim publishable anywhere**, or does this need the frequency work it currently forbids itself?
4. **Is it a contribution to show that a standard check cannot support the inference drawn from it?** (He et al., 2026) reported the depth phenomenon and the cosine decay first, and read the effect as depth on the warrant that edit success is stable across layers. §5.4 shows edit success cannot measure edit strength. Matching on belief displacement instead, the effect survives. Is that worth anything on its own?

8.2 What is open

Nine threads are open or active. They are listed because the useful question to a reader who knows the

field is not *"is this right"* but *"which of these is worth a month"* — and we cannot rank them from inside.

Table 10. Open threads, what each would decide, and what it costs. Ordered by

what we would spend next, which is a judgement we would like contradicted.

Two further threads are answered but worth naming because they cost us claims: killed an efficacy-metric critique in an hour by joining two files already on disk — and has since been **partly reversed** by a third file: ROME optimises `v*` until the target saturates, so post-edit log P(target) has an sd of **0.02** against the true answer's **5.68**, which makes ROME's ES *margin* rank-identical to the true answer's suppression ($\rho = 1.000$ exactly). The thread asked what ES fails to distinguish and answered with ES in disguise. The sharper critique — *ES measures true-answer suppression and nothing else* — is open again, with `||Δv||` as a quantity tracking propagation as well as ES does (-0.423 against $+0.425$) while nearly orthogonal to it (-0.276).Not yet a claim: correlational, below the effect floor its own ticket preregistered, and drawn from the 42 subjects that produced the observation. R-015 and cost six hours of compute to establish that a weak edit at depth was a hyperparameter artifact rather than a fact about depth.

The meta-result, and the reason §6 is the shape it is. Classifying this project's own 17 results by comparison shape found every narrowing came from **varying a dimension previously held fixed**, and none from adding arms to a dimension already varied. The three never varied anywhere were model, edit layer, and edited relation. Two of them are now §5.3 and §5.4. The third is .Prescriptively: before running a comparison, list what it holds fixed — model, layer, relation, template, pool, measurement method, decoding — and ask which has never been varied anywhere in the record. T-065

References

Every entry has had its title, authors, year and venue checked against the paper's own record. An entry that has not is drawn with a dotted rule and listed separately on Reading, so the mark appears only where something is wrong.

Appendix A. Why the coefficient is exactly 1 on a prefix-sharing prompt

§5.1 establishes this with the measurements around it. Stated on its own, without them:

Let the edit prompt be the token sequence $s_1 \dots s_m$, where s_m is the subject's last token, and let k^* be the layer-ℓ key read at position m. Let a probe be $s_1 \dots s_m\, t_1 \dots t_n$ — the same prefix, then any continuation.

Under causal attention the hidden state at position m is a function only of positions $1 \dots m$. The continuation occupies positions $m+1 \dots m+n$ and cannot enter it. So the probe's key at position m is $k = k^*$ exactly, and the update coefficient is

$$c = \frac{k^* \cdot u}{u \cdot k^*} = 1$$

The cancellation is independent of u, so it holds for any whitening covariance and at any layer, and independent of $t_1 \dots t_n$, so it holds whatever the probe goes on to ask. The empirical content is entirely in the premise — that the probe really does share the prefix through s_m — which is what §5.1 and §5.3 measured, across a 68× parameter range.

The tokenizer is the one place the premise can fail quietly: a leading space or a different casing produces a different token sequence and therefore a different key, with no error raised anywhere.

Appendix B. Provenance of every number

This project's provenance is not a table, because it outgrew one. It is the evidence ledger — 56 decisions, 33 findings and 85 threads, generated from the repository so it cannot drift from it, with falsification stated before each run rather than after it.

Every " marker in the sections above links into it. A number on this page that cannot be followed to a dated entry there is a defect, and finding one is a favour.

The ledger is deliberately the whole list rather than the ten claims this paper defends. Most entries went nowhere. A record that keeps only the entries that worked is not a record.

Appendix C. Readiness

What a venue asks for, against what exists today. A checklist for the author, not a claim about the work.

#	Claim, and what would refute it	value	status
F-01	Nobody probes the grounds that made an edited fact hold in the first place. **Reinstated if** RippleEdits' published test-type definitions turn out to probe only an edited fact's consequences. They do not: Logical Generalization names inverse and symmetric relations, which is a backward probe under another name. What survives is the narrower claim — unclaimed *for grounds against a decidable oracle*. R-005a		narrowed
F-02	An edit leaves its own grounds standing, and the resulting belief state is contradictory. **Reinstated if** a grounds set is found for which no satisfying world exists at all. Measured the other way: the grounds are evidential, so a world satisfying both the edit and the surviving grounds usually exists. What an edit leaves behind is improbable, not impossible.		withdrawn
F-03	A fact's presence in CounterFact is evidence that the model possesses it. **Reinstated if** any item's presence in CounterFact can be shown to depend on a reading taken from the model under test. The construction is model-independent, so membership says nothing about what any particular model holds. D-002		withdrawn
F-04	The `outer` leg lags because its answer pool is concentrated. **Reinstated if** the deficit survives with those two answer strings excluded. It does not. The original reasoning inferred a mechanism from a marginal count without ever joining the held flag against the answer string; the join dissolves it.	2 strings	withdrawn
F-05	The surface-form result is ours. **Reinstated only** by a reading of showing it does not cover the effect measured here. This withdrawal is a fact about the literature, not about the measurement — the measurement stands and the novelty does not. R-010		withdrawn
F-06	Knowledge status crossed with editability is unclaimed. **Reinstated only** by a reading of showing the pairing it reports is not this one. Same shape as the row above, and found the same way — by searching the literature after designing against it rather than before. R-010		withdrawn
F-07	The model revises the defeasible premise the edited fact rested on. **Reinstated if** an edit licensing no inference lands in the target country measurably less often than one that does. Both arms landed at **28 of 42**, identical, with subject and target country held fixed. What moves is the vector, not the inference.	67% vs 67%	withdrawn
F-08	Leakage onto the subject is structural in ROME rather than incidental. **Reinstated unconditionally if** leakage holds its magnitude as the edited layer is varied. It does not — reordering helps at depth — so what survives is *structural given the layer*, which is a claim about a configuration rather than about the method.		narrowed
F-09	Four constraints in nnsight bound what can be measured on a remote model. **Reinstated if** the three retired constraints reproduce against a healthy backend. They were a flaky one. The fourth is real, and it is the worse kind, because it fails without saying so. ·	1 of 4	narrowed
F-10	A probe sharing the edit prompt's prefix receives the delta in full — at any layer, under any covariance. **Analytic.** Refuted by a derivation error, or by its premise — prefix sharing — failing; the premise is what was measured. Across 16 chains the two groups do not overlap: the highest different-subject coefficient is **0.153**, the lowest same-subject **0.483**, and the different-subject floor sits at **0.082**. E-016 ·	1.000	standing
F-11	How much of an edit arrives predicts how far it travels. **Reinstated if** the estimate and measured movement correlate above $\rho = 0.3$ with delivery pinned at 1.000. Held at exactly 1.000, reach still varies **2.96×** — and the pinning is worst precisely across the probes whose outcomes differ most. This was never asserted here; it was checked because could not be answered without it, and answering it cost the strongest available reading of our own mechanism.	2.96×	withdrawn
F-12	An edit suppresses everything about the subject down to a common floor. **Reinstated if** the floor appears in a majority of arms on a fresh model-relation sweep. All six arms completed and it appears in **one**. What generalises instead is type matching — the probe whose answer type matches the injected value moves most, in 6 of 6.	1 of 6	withdrawn

Table 1: Table 5. Every claim this project made, what became of it, and what would change that.

ruled out	by	what it closes
Blast radius can be predicted from how much of the edit arrives		Delivery held at exactly 1.000 still gives **2.96×** variation in reach. The cheap-predictor route fails, and the measurement says why.
An edit suppresses everything about the subject uniformly		The floor is **1 of 6** arms. Two effects become one effect and one special case.
ROME's efficacy score misses what displacement catches	· *partly reinstated* R-015 · *narrowed*	The two predict propagation equally because they are **the same quantity** — ROME saturates its target, making the margin rank-identical to true-answer suppression ($\rho = 1.000$). So the thread answered its own question with the metric in disguise. A quantity the margin genuinely misses exists (`||Δv||`, -0.423 against $+0.425$, orthogonal at -0.276) but is **substantial in only one of seven configurations**. The critique is open again and much narrower than first drafted.
Different subjects are near-orthogonal at the subject position	·	A Llama-3.1-8B fact, not a ROME one, and architectural rather than an artifact of unfamiliar names.
The floor dips where ROME edits		Minima sit at different depths in different models, and in neither at the edited layer.
Backward probing is unexplored · the surface-form result is ours · knowledge-status × editability is unclaimed	R-001 · R-010	Three framings retired against existing literature, two of them after we had designed around them.

thread	the question	what a result would settle	cost
	What sets how far an edit travels?	Seven candidates eliminated at a pre-registered floor; the best pairing of the survivors is bounded at **0.534** in closed form. Nothing explains more than ~18% alone or ~28% together.	unknown — and now *searched*, which is the more useful form of unknown
	Is GPT-J's higher floor real, or unfamiliarity?	Whether §5.3's narrowing of the floor to a Llama fact is confirmed or an artifact of running Llama-gated subjects on another model	hours
	Does the key still reorient when context cannot reach it?	answered : attention never writes more of the divergence than the MLP at any depth (largest share 1.03), and a matched-length *unrelated* prefix still produces ~60% of a related one's divergence mid-network. But the MLP's input comes through attention, so only an ablation separates them.	a day
	Is the floor-plus-displacement structure a *birthplace* fact?	§5.5 is measured on one relation. Editing occupation instead inverts the prediction; a clean crossover would establish the mechanism.	a day
	Why is the different-subject floor lowest near the edit layer?	Both models' floors dip near depth fraction 0.15, which is roughly where ROME edits. Either a connection to ROME's own causal-tracing layer choice, or a coincidence at n≈12.	a day
	CounterFact runs no possession check	Its premise was reversed by D-002; it needs re-reading rather than running	reading

	requirement	state today	where
yes	A result that could have come out the other way	The work-country control relocates birthplace at an identical rate, paired difference +0.0 pp, McNemar p = 1.000 — and it was run to break our own reading	§5.2
yes	Measured on materials we did not choose	The same measurement run on RippleEdits' and MQuAKE's own edits and prompts	§5.6
yes	Every withdrawn claim kept on the page	Twelve claims, nine withdrawn or narrowed, each with what changed it	§6
yes	Reproduction path	Every run levelled and committed, including the failures; falsification stated before each	§4
no	A rate rather than an existence claim	n = 42, one model, one relation family, one layer. No rate is defensible and the artifacts are written to avoid implying one	§7.1
no	What sets propagation magnitude	Seven predictors eliminated against a pre-registered floor; nothing explains more than about 28% together. Searched, which is the more useful form of unknown	§8.2
no	Read by someone outside the project	Every finding produced and reviewed by the same two parties	—

Table 2: Readiness against a short-paper submission.